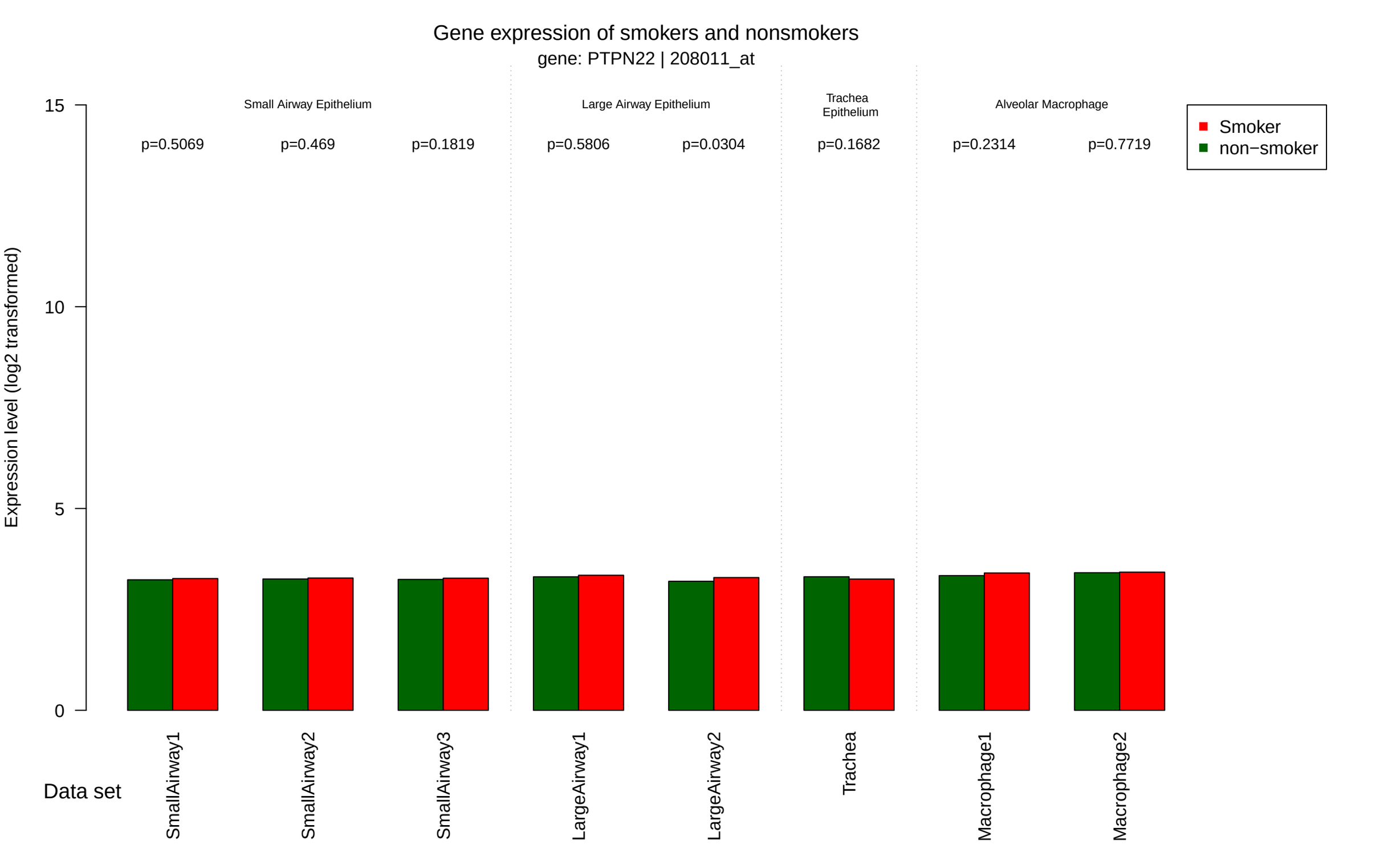
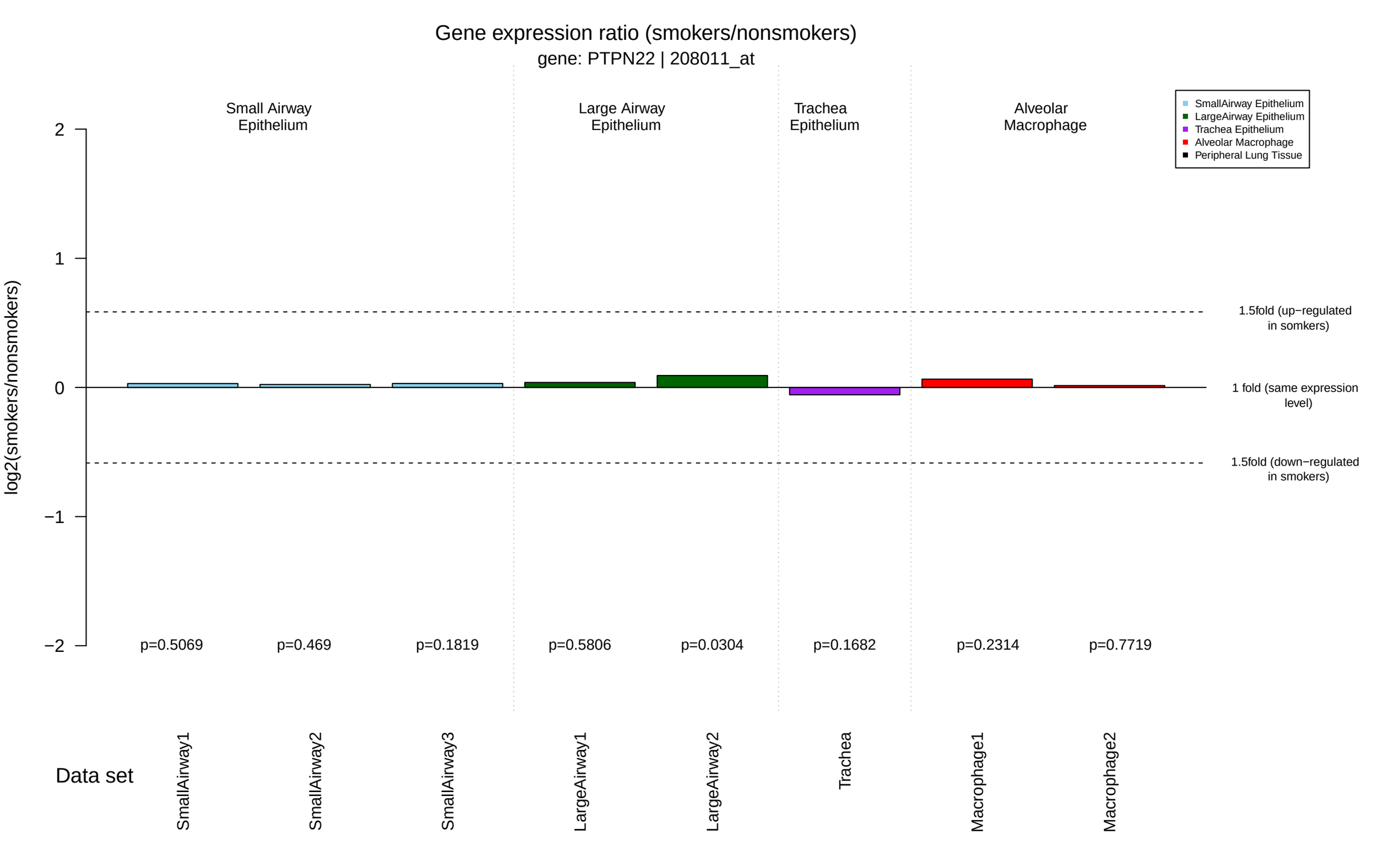
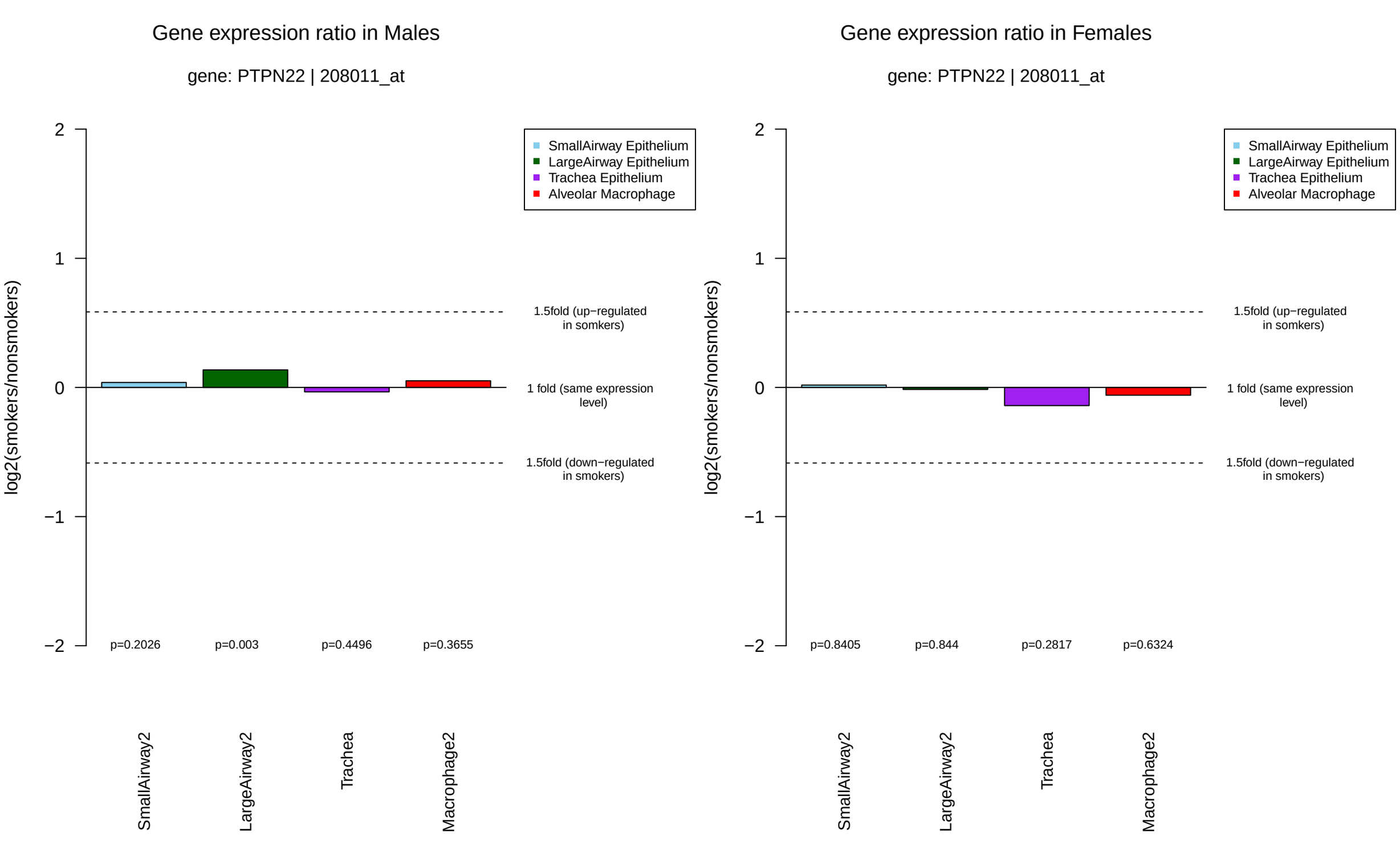
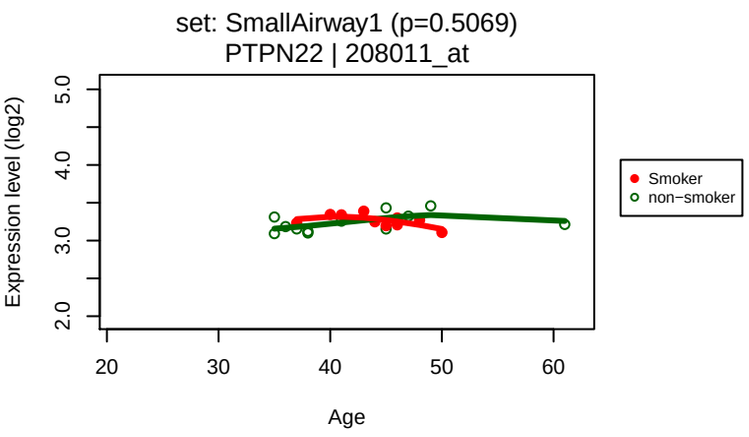
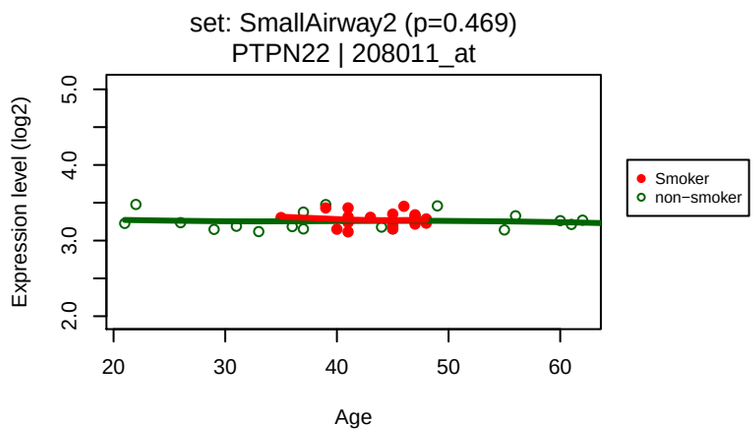
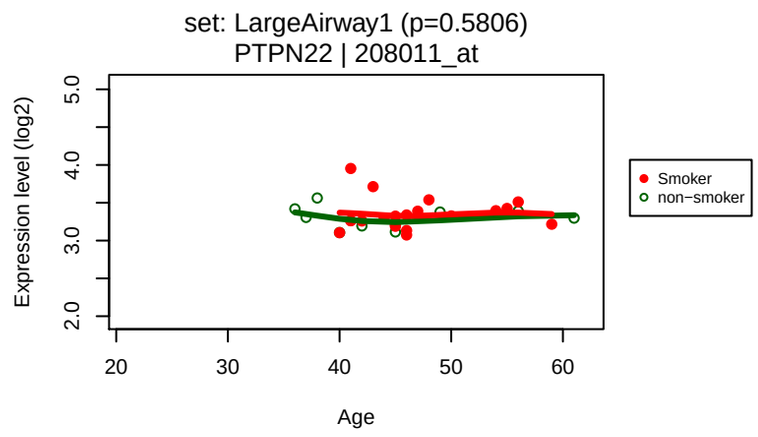

Gene expression of smokers and nonsmokers
gene: PTPN22 | 208011_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5069
p=0.469
p=0.1819
p=0.5806
p=0.0304
p=0.1682
p=0.2314
p=0.7719
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

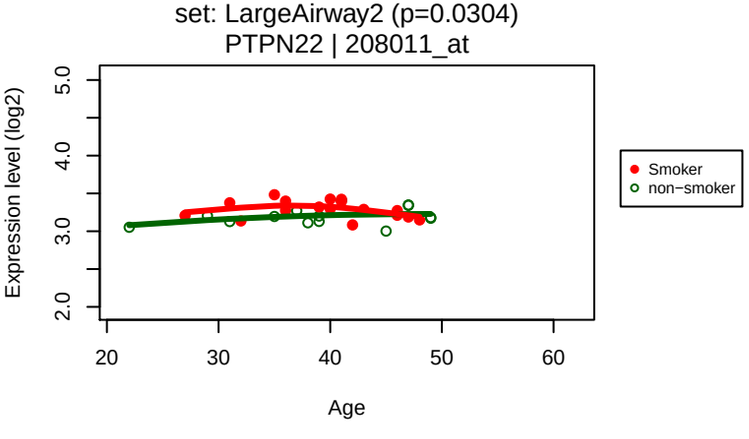

Gene expression ratio (smokers/nonsmokers)
gene: PTPN22 | 208011_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5069
p=0.469
p=0.1819
p=0.5806
p=0.0304
p=0.1682
p=0.2314
p=0.7719
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

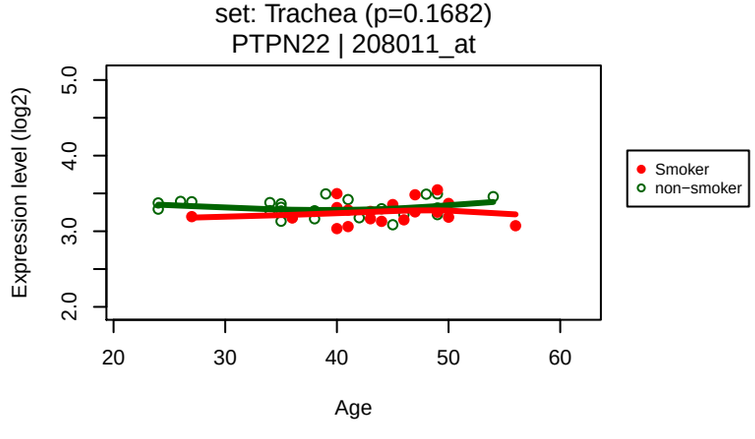

Gene expression ratio in Males
gene: PTPN22 | 208011_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2026
p=0.003
p=0.4496
p=0.3655
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTPN22 | 208011_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8405
p=0.844
p=0.2817
p=0.6324
SmallAirway2
LargeAirway2
Trachea
Macrophage2

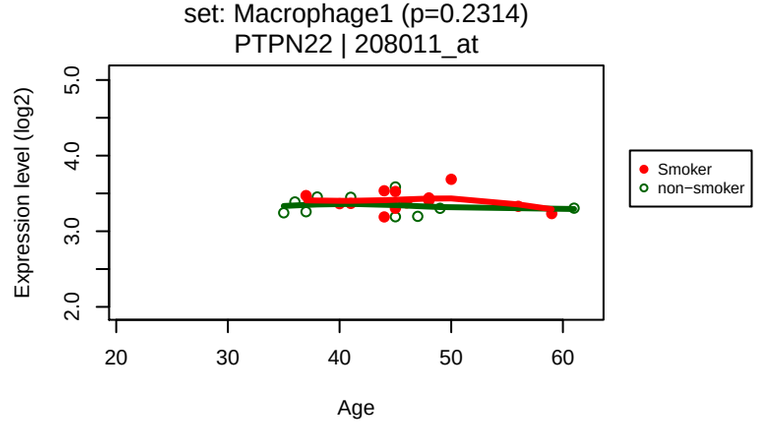

set: SmallAirway1 (p=0.5069)
PTPN22 | 208011_at
Expression level (log2)
Age
Smoker
non−smoker

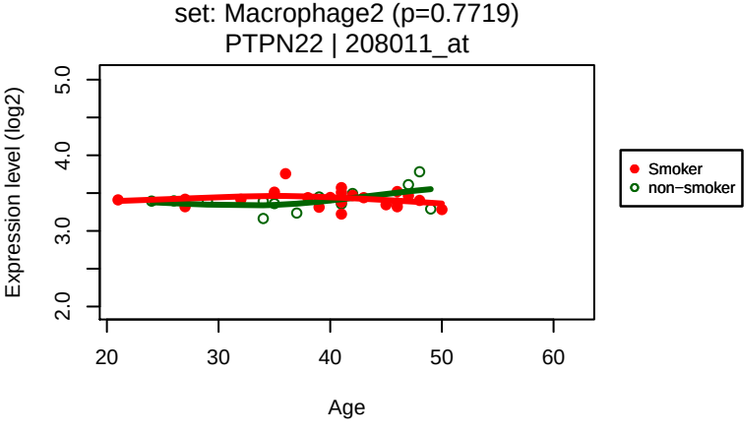

set: SmallAirway2 (p=0.469)
PTPN22 | 208011_at
Expression level (log2)
Age
Smoker
non−smoker

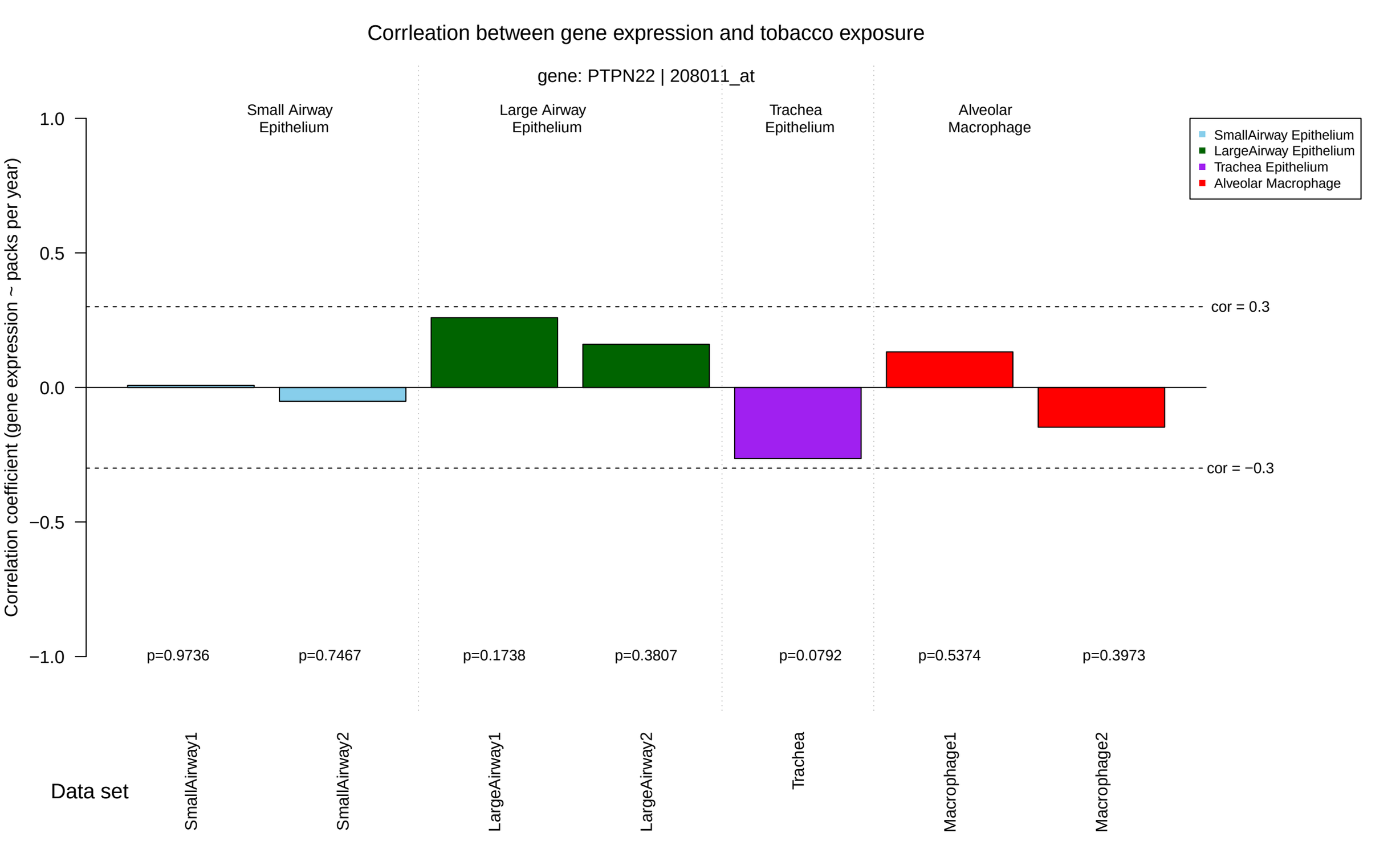

set: LargeAirway1 (p=0.5806)
PTPN22 | 208011_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0304)
PTPN22 | 208011_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1682)
PTPN22 | 208011_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2314)
PTPN22 | 208011_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7719)
PTPN22 | 208011_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTPN22 | 208011_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9736
p=0.7467
p=0.1738
p=0.3807
p=0.0792
p=0.5374
p=0.3973
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2